

(a) INPUTS

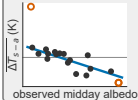
Observations

Urban-PLUMBER
19 records · 18 towers · 16 cities
30-min $LW_{\uparrow}$, $LW_{\downarrow}$, T_a
albedo · geometry

Simulations

TEB 4.1.2 · CLM-Urban (CLM5.0)
offline · shared forcing
default materials

Observed association



albedo vs. mean ΔT_{s-a}
 $r = -0.77$ (core 16)
-2.7 K per 0.1 albedo
association only

(b) COMMON PROCESSING

Matched nocturnal data

observed & simulated $LW_{\uparrow}$
one sampling mask

Radiometric diagnostic

$$T_s = \left[\frac{LW_{\uparrow} - (1 - \varepsilon) LW_{\downarrow}}{\varepsilon \sigma} \right]^{1/4}$$

$$\varepsilon = 0.95$$

Record-mean offset

$$\overline{\Delta T_{s-a}} = \overline{T_s} - \overline{T_a}$$

19 means per dataset

reference

(c) EVALUATION

1 Temporal skill

$LW_{\uparrow}$ time series
 r · cRMSE · agreement



2 Bias & benchmarks

mean bias · MAE
REG2 · KM3
leave-one-city-out



3 Cross-site gradient

spread s_m/s_o · ordering r
slope $\hat{\beta} = r(s_m/s_o)$
1:1 reference



(d) TARGETED ROBUSTNESS CHECKS

Observation uncertainty

random · placement
albedo-aligned · 208 cells

Sampling

city clusters · seasons
epochs · record sets

Model settings

materials · albedo
spin-up · structural probe